

Fair Collaborative Classification via Synthetic Bilevel Optimization and Universum Pseudo-Positives

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Abstract

We study collaborative classification with a binary sensitive attribute under privacy constraints and propose a practical framework that achieves *Equal Opportunity* (EO) fairness without sharing raw data. Each client constructs a per-round training payload composed of (i) a labeled *synthetic* minibatch whose (attribute, label) composition is fixed by joint stratification and small positive floors, and (ii) *Universum* pseudo-positive points that counter label imbalance while preserving privacy. To learn these payloads, we formulate a *features-only* bilevel program: the inner learner fits a model on synthetic plus Universum data, while the outer objective—evaluated *only* on the client’s original minibatch—enforces EO via a conditional-covariance surrogate. We solve the constrained outer problem with an *Augmented Lagrangian* method and obtain hypergradients by implicit differentiation using Hessian–vector products, yielding an efficient, fully differentiable procedure. Universum points prevent balance the inner class mix, and enable *margin shaping* beyond what uniform class reweighting provides. Only synthetic and Universum features are shared; originals never leave the client. The result is a privacy-preserving, minibatch-friendly pipeline that (i) provides clear mini to global EO guarantees, (ii) stabilizes minority recall through Universum pseudo-positives, and (iii) scales to expressive nonlinear models with lightweight per-round computation.

1 Introduction

In collaborative (federated) classification, private datasets are distributed across many clients that differ in covariates, class priors, and group composition. When a sensitive attribute is present [1], naively aggregating such heterogeneous sources can yield globally unfair models, while privacy constraints forbid sharing raw examples. We seek a protocol that *evaluates fairness on each client’s original distribution* yet trains on *shareable* data, so that a single global classifier remains accurate and fair at population scale.

We adopt EO as our fairness notion: among qualified individuals, the likelihood of a positive prediction should not depend on group membership. Let $A \in \{0, 1\}$ denote a binary sensitive attribute, $Y \in \{0, 1\}$ the ground-truth label, and $\hat{Y}$ the prediction. EO requires equality of true positive rates,

$$\mathbb{P}(\hat{Y} = 1 \mid Y = 1, A = 0) = \mathbb{P}(\hat{Y} = 1 \mid Y = 1, A = 1), \quad (1)$$

which protects those who deserve a favorable decision by conditioning on $Y = 1$.

We propose a *client-centric bilevel framework* that enforces EO locally while preserving privacy and robustness to imbalance. In each round, a client draws an *original* minibatch via joint stratification over (A, Y) with small per-group $Y=1$ floors to stabilize EO estimation; *accuracy and EO are evaluated only on this original minibatch*. From it, the client constructs a labeled *synthetic*

minibatch that matches the (A, Y) composition (labels fixed; features optimized) and augments it with *Universum* pseudo-positives ($Y=1$) to counter label imbalance and bolster minority recall. The inner model is trained on the synthetic+Universum payload, while an augmented-Lagrangian controller enforces a minibatch-level EO surrogate; only synthetic plus Universum *features* (never raw data) are sent to the server for aggregation into a single global update. By repeating this procedure, client-level fairness constraints and stratified sampling guide the global model toward population-level EO, yielding a practical, minibatch-friendly pipeline that preserves privacy, handles heterogeneity and class imbalance, and scales to modern training workloads.

2 Iterative Bilevel Framework for Fair Synthetic and Universum Data Construction

In collaborative classification, each client k holds an *original* dataset $\mathcal{D}_k^o = \{(x_i^o, a_i^o, y_i^o)\}_{i=1}^{N_k}$, where $x_i^o \in \mathbb{R}^d$ is the feature vector, $a_i^o \in \{0, 1\}$ is a binary sensitive attribute, and $y_i^o \in \{0, 1\}$ is a binary label that may be imbalanced. Our goal is an iterative (per-round) bilevel procedure that trains on stratified minibatches and synthetic payloads to maintain accuracy and Equal Opportunity (EO) fairness across clients. For the core notation table, see Table:

Symbol	Type/Range	Meaning
$B_{k,t}$	set	<i>Original</i> minibatch at round t (drawn without replacement within the round).
$ B_{k,t} $	$\mathbb{N}$	Size of original minibatch.
$\Delta_{k,t}$	$\mathbb{N}$	Original-batch label imbalance, $\Delta_{k,t} = n_{k,t}^{(0)} - n_{k,t}^{(1)} (\geq 0)$.
$\mathcal{D}_{k,t}^s$	set	<i>Labeled synthetic</i> minibatch (labels/attributes fixed; features optimized).
$ \mathcal{D}_{k,t}^s $	$\mathbb{N}$	Size of labeled synthetic minibatch, with $ \mathcal{D}_{k,t}^s \leq B_{k,t} $.
$\Delta_{k,t}^s$	$\mathbb{N}$	Synthetic-batch label imbalance induced by quotas.
$\mathcal{U}_{k,t}^{(Y)}$	set	Universum pseudo-positives ($y=1$), split evenly across $a \in \{0, 1\}$.
$ \mathcal{U}_{k,t}^{(Y)} $	$\mathbb{N}$	Universum size.
$\tilde{S}_{k,t}$	set	Payload sent to server: $\tilde{S}_{k,t} = \mathcal{D}_{k,t}^s \cup \mathcal{U}_{k,t}^{(Y)}$.

Table 1: Core symbols used in the per-round construction.

At round t , client k : (i) draws an original minibatch $B_{k,t}$ via joint (a, y) -stratification with per-group $Y=1$ floors and records the label imbalance $\Delta_{k,t} = n_{k,t}^{(0)} - n_{k,t}^{(1)}$; (ii) chooses a synthetic size $|\mathcal{D}_{k,t}^s|$ such that $|\mathcal{D}_{k,t}^s| \leq |B_{k,t}|$ and forms the labeled synthetic minibatch $\mathcal{D}_{k,t}^s = \{(x_j^s, a_j^s, y_j^s)\}$ by rescaling the (a, y) quotas from $B_{k,t}$ (labels/attributes fixed; features $\{x_j^s\}$ are decision variables), letting $\Delta_{k,t}^s$ denote the induced synthetic imbalance; (iii) creates a Universum set of pseudo-positives $\mathcal{U}_{k,t}^{(Y)} = \{(x_u^s, a_u^s, 1)\}$ with $|\mathcal{U}_{k,t}^{(Y)}| = \min\{\Delta_{k,t}^s, |\mathcal{D}_{k,t}^s|, \Delta_{k,t}\}$ and an even split over $a \in \{0, 1\}$ (features $\{x_u^s\}$ are decision variables). We sample *without replacement* within each $B_{k,t}$; across rounds, *reuse is allowed* (the same original point may reappear).

The client transmits the synthetic payload $\tilde{S}_{k,t} = \mathcal{D}_{k,t}^s \cup \mathcal{U}_{k,t}^{(Y)}$ to the server; labels and attributes in $\tilde{S}_{k,t}$ are fixed templates and only features are optimized. Crucially, the outer objective and EO constraint are computed *only* on the original minibatch $B_{k,t}$. See Appendix 4.2 for detailed construction rules (quota renormalization, randomized rounding, initialization, projections) and ablations.

Definition 1 (Per-round losses and EO surrogate for client k at iteration t). Let ℓ be a convex classification loss and f_θ a parametric score. Denote the feature dimension by d (so $d+1$ includes the sensitive attribute A). Let ζ be a reference parameter (e.g., the previous global model). We define:

(i) Generic base loss for any dataset $\mathcal{D}_{k,t} = \{(x, a, y)\}$, reference ζ , and regularization weight $\lambda_\theta \geq 0$:

$$\mathcal{L}_{k,t}(\theta; \mathcal{D}_{k,t}, \zeta, \lambda_\theta) := \frac{1}{|\mathcal{D}_{k,t}|} \sum_{(x,a,y) \in \mathcal{D}_{k,t}} \ell(y, f_\theta(x, a)) + \frac{\lambda_\theta}{2(d+1)^2} \|\theta - \zeta\|_2^2. \quad (2)$$

(ii) Universum-only loss (pseudo-positives, balanced in S):

$$\mathcal{L}_{k,t}^U(\theta; \mathcal{U}_{k,t}^{(Y)}) := \begin{cases} \frac{1}{|\mathcal{U}_{k,t}^{(Y)}|} \sum_{(x_u^s, s_u^s, 1) \in \mathcal{U}_{k,t}^{(Y)}} \ell(1, f_\theta(x_u^s, s_u^s)), & \Delta_{k,t}^s > 0, \\ 0, & \Delta_{k,t}^s = 0, \end{cases} \quad (3)$$

where $|\mathcal{U}_{k,t}^{(Y)}| = \min\{\Delta_{k,t}^s, |\mathcal{D}_{k,t}^s|, \Delta_{k,t}\}$ and the sensitive labels in $\mathcal{U}_{k,t}^{(Y)}$ are split as evenly as possible across $a \in \{0, 1\}$.

(iii) EO surrogate (batch-level, computed on original positives): with $B_{k,t}^{(+)} = \{(x^o, s^o, 1) \in B_{k,t}\}$ and

$$\bar{s}_{k,t}^{(+)} := \frac{1}{|B_{k,t}^{(+)}|} \sum_{(x^o, s^o, 1) \in B_{k,t}^{(+)}} s^o,$$

define

$$g_{\text{EO}}^{k,t}(\theta) := \frac{1}{|B_{k,t}^{(+)}|} \sum_{(x^o, s^o, 1) \in B_{k,t}^{(+)}} (s^o - \bar{s}_{k,t}^{(+)}) f_\theta(x^o, s^o). \quad (4)$$

Given the global update from round $t-1$, set $\zeta_{\text{in}} = \theta_{t-1}^{\text{glob}}$. Let $B_{k,t}$ be the original stratified minibatch with per-group $Y=1$ floors, and let $\Delta_{k,t} = n_{k,t}^{(0)} - n_{k,t}^{(1)} \geq 0$. Select $|\mathcal{D}_{k,t}^s|$ such that $|\mathcal{D}_{k,t}^s| \leq |B_{k,t}|$, and using stratification on (a, y) , ensure that $\{A_i, Y_i\}$ has the same distribution. we construct $\mathcal{D}_{k,t}^s = \{(x_j^s, a_j^s, y_j^s)\}_{j=1}^{|\mathcal{D}_{k,t}^s|}$, where features $\{x_j^s\}$ are decision variables, and $\mathcal{U}_{k,t}^{(Y)} = \{(x_u^s, s_u^s, 1)\}_{u=1}^{|\mathcal{U}_{k,t}^{(Y)}|}$ (split as evenly as possible across S ; features $\{x_u^s\}$ are decision variables). Then

$$\begin{aligned} \min_{\{x_j^s\}_{j=1}^{|\mathcal{D}_{k,t}^s|}, \{x_u^s\}_{u=1}^{|\mathcal{U}_{k,t}^{(Y)}|}} \quad & \underbrace{\mathcal{L}_{k,t}(\theta^*; B_{k,t}, 0, \lambda_{\theta, \text{out}})}_{\text{outer: original minibatch + outer ridge}} \\ \text{s.t.} \quad & \underbrace{g_{\text{EO}}^{k,t}(\theta^*)}_{\text{EO on originals}} \leq \varepsilon_{\text{EO}}, \\ & \theta^* \in \arg \min_{\vartheta} \left\{ \underbrace{\mathcal{L}_{k,t}(\vartheta; \mathcal{D}_{k,t}^s, \zeta_{\text{in}}, \lambda_{\theta, \text{in}})}_{\text{inner base (prox to } \theta_{t-1}^{\text{glob}})} + \underbrace{\lambda_U \mathcal{L}_{k,t}^U(\vartheta; \mathcal{U}_{k,t}^{(Y)})}_{\text{Universum pseudo-positives}} \right\}. \end{aligned} \quad (5)$$

Only the features $\{x_j^s\}$ and $\{x_u^s\}$ are decision variables; (a_j^s, y_j^s) and $(s_u^s, 1)$ are fixed by stratification/balancing. In practice, the EO constraint is enforced via an Augmented Lagrangian on $B_{k,t}$.

We clarify why we juxtapose Universum pseudo-positives with the widely used alternative of class reweighting. Both aim to correct minority under-representation in the *training* signal, while our EO constraint is enforced on *original* data. Reweighting is a natural baseline because, in

expectation, adding suitably distributed Universum positives can mimic an increased positive weight; this provides readers a familiar lens for interpreting our inner objective. At the same time, the comparison highlights what we need beyond reweighting in collaborative, privacy-preserving settings: (i) a way to shape the *geometry* of the decision boundary by placing pseudo-positives near hard regions (margin shaping), (ii) explicit control over the *cardinality and balance* of added positives without leaking originals, and (iii) a design that keeps fairness accounting clean by separating inner payload shaping from EO evaluation on originals. Framing Universum through this reweighting lens thus motivates our choice: we recover the intuitive “more influence for qualified examples” effect *and* gain finer, geometry-aware control that better protects the minority class—setting the stage for our bilevel construction and augmented Lagrangian solver.

Our inner objective augments the synthetic labeled set $\mathcal{D}_{k,t}^s$ with Universum pseudo-positives $\mathcal{U}_{k,t}^{(Y)}$:

$$L_{\text{in}}(\vartheta) = \underbrace{\frac{1}{|\mathcal{D}_{k,t}^s|} \sum_{(x,a,y) \in \mathcal{D}_{k,t}^s} \ell(y, f_{\theta}(x, a))}_{\text{synthetic labeled}} + \lambda_U \underbrace{\frac{1}{|\mathcal{U}_{k,t}^{(Y)}|} \sum_{(x,a,1) \in \mathcal{U}_{k,t}^{(Y)}} \ell(1, f_{\theta}(x, a))}_{\text{Universum pseudo-positives}}.$$

If $\mathcal{U}_{k,t}^{(Y)}$ is sampled to mimic the positive class distribution $P(X | Y=1)$, then in expectation the Universum term acts like a *class-weight* on positives. Writing $m_1 = \#\{(x, a, 1) \in \mathcal{D}_{k,t}^s\}$ and $m_0 = \#\{(x, a, 0) \in \mathcal{D}_{k,t}^s\}$, we obtain the approximation

$$\mathbb{E}[L_{\text{in}}(\theta)] \approx \frac{1}{|\mathcal{D}_{k,t}^s|} \sum_{(x,a,0) \in \mathcal{D}_{k,t}^s} \ell(0, f_{\theta}) + \frac{w_p}{|\mathcal{D}_{k,t}^s|} \sum_{(x,a,1) \in \mathcal{D}_{k,t}^s} \ell(1, f_{\theta}), \quad w_p \approx 1 + \lambda_U \frac{|\mathcal{D}_{k,t}^s|}{m_1}.$$

Thus, when Universum points follow the positive manifold, adding them *implicitly reweights* the positive class, without introducing explicit per-example weights in the main exposition.

Universum pseudo-positives let us shape the decision boundary, not just adjust the class prior: by placing synthetic $Y=1$ examples near the margin or in under-represented regions, we obtain targeted gradients that lift minority recall with few added points—an effect uniform reweighting does not explicitly provide. Their cardinality and balance are directly controlled by $|\mathcal{U}_{k,t}^{(Y)}| = \min\{\Delta_{k,t}^s, |\mathcal{D}_{k,t}^s|, \Delta_{k,t}\}$, with an even split across the sensitive attribute, which prevents payload domination and keeps the inner class mix well-conditioned. Because Universum points are synthetic, they can match local feature statistics without revealing originals, aligning with our privacy model. Finally, EO is enforced and evaluated only on *original* data $B_{k,t}$; Universum affects only the inner optimization, so we avoid weighted-loss machinery while still amplifying the influence of qualified examples during training.

3 Augmented Lagrangian Approach to Bilevel Optimization

At client k and round t , the *inner* learner fits parameters θ on a payload that consists of a labeled *synthetic* minibatch $\mathcal{D}_{k,t}^s$ and a set of *Universum* pseudo-positives $\mathcal{U}_{k,t}^{(Y)}$ (labels fixed by templates; only features are optimized). The *outer* objective evaluates task loss on the *original* minibatch $B_{k,t}$ and enforces Equal Opportunity (EO) on $B_{k,t}$ via the minibatch-level surrogate $g_{\text{EO}}^{k,t}(\theta)$ (cf. Def./Eq. in the previous section). We solve the resulting constrained bilevel by an *Augmented Lagrangian* (AL): EO appears as a linear penalty with multiplier λ plus a quadratic penalty with weight ρ , while the bilevel coupling is handled by *implicit differentiation* (or, as a fallback, short unrolled inner steps).

3.1 Client-side Augmented Lagrangian Solver (one round)

Let $\zeta_{\text{in}} = \theta_{t-1}^{\text{glob}}$ be the reference (broadcast) model. The inner and outer objectives are

$$L_{\text{in}}(\theta) = \underbrace{\mathcal{L}_{k,t}(\theta; \mathcal{D}_{k,t}^s, \zeta_{\text{in}})}_{\text{synthetic labeled}} + \lambda_U \underbrace{\mathcal{L}_{k,t}^U(\theta; \mathcal{U}_{k,t}^{(Y)})}_{\text{Universum}}, \quad L_{\text{out}}(\theta) = \mathcal{L}_{k,t}(\theta; B_{k,t}, 0),$$

and the AL outer objective is

$$\Phi(\theta; \lambda, \rho) = L_{\text{out}}(\theta) + \lambda g_{\text{EO}}^{k,t}(\theta) + \frac{\rho}{2} g_{\text{EO}}^{k,t}(\theta)^2. \quad (6)$$

Algorithm 1 Client-side AL solver for round t (features-only outer; EO on originals)

Input: Original minibatch $B_{k,t}$; templates $\mathcal{D}_{k,t}^s = \{(x_j^s, a_j^s, y_j^s)\}$, $\mathcal{U}_{k,t}^{(Y)} = \{(x_u^s, a_u^s, 1)\}$ with sizes $|\mathcal{D}_{k,t}^s| \leq |B_{k,t}|$ and $|\mathcal{U}_{k,t}^{(Y)}| = \min\{\Delta_{k,t}^s, |\mathcal{D}_{k,t}^s|, \Delta_{k,t}\}$. Reference $\zeta_{\text{in}} = \theta_{t-1}^{\text{glob}}$; hyperparams $\lambda_{\theta, \text{in}}, \lambda_{\theta, \text{out}}, \lambda_U, \rho > 0$, step sizes η_θ, η_x , inner steps K , outer steps J .

- 1: Initialize $\theta \leftarrow \zeta_{\text{in}}$; $\lambda \leftarrow 0$; initialize features $\{x_j^s\}, \{x_u^s\}$ (e.g., moment-matched) and project to $\|x\| \leq R$.
- 2: **for** $j = 1$ to J **do** ▷ Outer iterations
- 3: **Inner fit (warm-start):** run K steps of SGD/Adam on $L_{\text{in}}(\theta)$ to get θ^* .
- 4: **EO & outer loss on originals:** compute $g \leftarrow g_{\text{EO}}^{k,t}(\theta^*)$ on $B_{k,t}^{(+)}$ and $\ell_{\text{out}} \leftarrow L_{\text{out}}(\theta^*)$.
- 5: **AL value:** $\Phi \leftarrow \ell_{\text{out}} + \lambda g + \frac{\rho}{2} g^2$.
- 6: **Hypergradient (implicit):** set $v \leftarrow \nabla_\theta \Phi(\theta^*)$; solve $Hh = v$ approximately by CG (few iters), where $H = \nabla_{\theta\theta}^2 L_{\text{in}}(\theta^*)$ using Hessian-vector products.
- 7: **Feature gradients:** for each synthetic plus Universum feature block $x \in \{x_j^s\} \cup \{x_u^s\}$,

$$\nabla_x \Phi = -\left(\frac{\partial^2 L_{\text{in}}}{\partial x \partial \theta}(\theta^*)\right)^\top h \quad (\text{Jacobian-vector via autodiff});$$

note L_{out} depends on x only through θ^* .

- 8: **Outer updates (with projection):** $x \leftarrow \Pi_{\|x\| \leq R}(x - \eta_x \nabla_x \Phi)$ for all $x \in \{x_j^s\} \cup \{x_u^s\}$.
- 9: **Multiplier update:** $\lambda \leftarrow \lambda + \rho g$; **penalty schedule:** if $|g| > \varepsilon_{\text{EO}}$ for M outer steps, set $\rho \leftarrow \gamma \rho$ (e.g., $\gamma = 2$).
- 10: **Stopping (optional):** if $|g| \leq \varepsilon_{\text{EO}}$ and $|\Delta \ell_{\text{out}}|/\ell_{\text{out}} < \tau$, **break**.
- 11: **end for**

Output: Payload $\tilde{S}_{k,t} = \mathcal{D}_{k,t}^s \cup \mathcal{U}_{k,t}^{(Y)}$ with optimized features; originals $B_{k,t}$ never leave the client.

The implicit step uses the standard formula

$$\nabla_x \Phi = -\left(\nabla_{x\theta}^2 L_{\text{in}}(\theta^*)\right)^\top H^{-1} \nabla_\theta \Phi(\theta^*),$$

computed without forming H explicitly via CG and Hessian-vector products. As a memory-light alternative, one may *unroll* the K inner updates and backpropagate; with small K both approaches are effective.

3.2 Server-side Aggregation and Global Update

At the end of round t , the server collects $\{\tilde{S}_{k,t}\}_k$ and trains a global classifier on the union $\bigcup_k \tilde{S}_{k,t}$ (e.g., ridge-regularized ERM with early stopping). The resulting model θ_t^{glob} is broadcast and used

as the next-round reference ζ_{in} . This preserves privacy (only synthetic plus Universum features are shared) and maintains a clean separation: EO is always measured on clients’ originals.

References

- [1] Chia-Yuan Wu, Frank E Curtis, and Daniel P Robinson. “Using Synthetic Data to Mitigate Unfairness and Preserve Privacy in Collaborative Machine Learning”. In: *arXiv preprint arXiv:2409.09532* (2024).

4 Appendix

This appendix has two parts. The first collects theoretical results that underpin our methodology (assumptions, propositions, and proofs). The second details the per-round minibatch design and includes a worked example.

Clarification (Meaning of B_t). In the collaborative setting, each client k draws at round t a local minibatch $B_{k,t} \subset \mathcal{D}_k^o$ from its original dataset. For theoretical analysis, we view the union of all local batches as a *conceptual global batch*

$$B_t := \bigcup_{k=1}^K B_{k,t}.$$

The event that a sample “belongs to B_t ” means that this sample—regardless of which client owns it—has been selected for participation in round t . Probabilities such as $P(B_t \mid S = s, Y = y)$ therefore denote the likelihood that a randomly chosen individual from subgroup $(S = s, Y = y)$ is included (through its client’s draw) in the global batch for that round. In practice, the server never observes raw B_t ; it is a *theoretical abstraction* used to analyze how minibatch-level fairness aggregates to the global population. Independence $P(B_t \mid S = s, Y = y) = P(B_t)$ is ensured by joint (S, Y) -stratified sampling within each client.

4.1 Theoretical Results

Lemma 1 (Batch membership independence under joint stratification). *Let $B_{k,t} = \bigcup_{(a,y)} B_{k,t}^{(a,y)}$ be built with quotas $q_{k,t}^{(a,y)}$ such that $q_{k,t}^{(a,y)} = |B_{k,t}| \pi_k^{(a,y)}$ (exactly or in expectation via randomized rounding). For a uniformly random original index U ,*

$$\mathbb{P}(U \in B_{k,t} \mid S_U = s, Y_U = y) = \frac{|B_{k,t}|}{N_k} = \mathbb{P}(U \in B_{k,t}).$$

Proof sketch. Within (a, y) , sampling $q_{k,t}^{(a,y)}$ from $N_{k,s,y}^o$ yields $q_{k,t}^{(a,y)} / N_{k,s,y}^o = |B_{k,t}| / N_k$. In cases where within-batch replacement is used for a tiny stratum, the identity holds in expectation (with quotas satisfied in expectation).

Lemma 2 (Per-batch qualified floors stabilize EO). *If each minibatch enforces $q_{k,t}^{(a,1)} \geq c_{\min,k}$ for $a \in \{0, 1\}$, then $B_{k,t}^{(+)}$ contains at least $c_{\min,k}$ positives for each S , so $g_{\text{EO}}^{k,t}(\theta)$ is always defined and its variance is controlled (denominator bounded away from zero), yielding stable AL updates.*

Theorem 1 (Mini-to-Global EO with variable original minibatch sizes). *Let $\Delta_{k,t}(\theta, \tau)$ be the TPR gap on the original minibatch $B_{k,t}$. Then*

$$\mathbb{P}(\hat{Y}=1 \mid Y=1, S=0) - \mathbb{P}(\hat{Y}=1 \mid Y=1, S=1) = \sum_k \sum_t \underbrace{\frac{|B_{k,t}|}{\sum_j N_j}}_{w_{k,t}^{\text{global}}} \Delta_{k,t}(\theta, \tau).$$

Within a fixed client k , the client-level gap equals $\sum_t \frac{|B_{k,t}|}{N_k} \Delta_{k,t}(\theta, \tau)$. If, in addition, the conditional densities of f_θ near τ are bounded on the $Y=1$ slice of each $B_{k,t}$, then

$$|\Delta_{k,t}(\theta, \tau)| \leq \frac{\lambda_{\text{dens}}^{(k,t)}}{\bar{p}_{k,t}(1 - \bar{p}_{k,t})} |g_{\text{EO}}^{k,t}(\theta)|, \quad \bar{p}_{k,t} = \mathbb{P}(S=1 \mid Y=1, B_{k,t}),$$

and hence the global EO gap is bounded by the corresponding weighted sum of $|g_{\text{EO}}^{k,t}(\theta)|$ terms. Proof sketch. Apply the law of total probability with Lemma 1 to express TPRs as weighted sums over minibatches; then use a mean-value/density argument to relate the TPR gap to the surrogate on each minibatch, and convexity of absolute value with weights.

Remark 1 (No-reuse across rounds (optional)). If one forbids reuse of original points across all T rounds, then the quotas must satisfy the global constraints $\sum_{t=1}^T q_{k,t}^{(a,y)} \leq N_{k,s,y}^o$ for each stratum, which implies a cap on the number of rounds:

$$T \leq T_{\text{no-reuse},k}^{\max} := \min_{(a,y)} \left\lfloor \frac{N_{k,s,y}^o}{q_{\text{floor},k}^{(a,y)}} \right\rfloor, \quad q_{\text{floor},k}^{(a,1)} = \max\{\text{round}(\pi_k^{(a,1)} |B_{k,t}|), c_{\min,k}\}.$$

Our method adopts the standard regime that allows reuse across rounds, so this cap is not binding.

Communication payload. At round t , client k transmits the *synthetic payload* $\tilde{S}_{k,t} = \mathcal{D}_{k,t}^s \cup \mathcal{U}_{k,t}^{(Y)}$ with size $|\tilde{S}_{k,t}| = |\mathcal{D}_{k,t}^s| + |\mathcal{U}_{k,t}^{(Y)}|$. The outer objective $\ell_{\text{task}}(\theta; B_{k,t})$ and EO surrogate $g_{\text{EO}}^{k,t}(\theta)$ are evaluated only on the *original* minibatch $B_{k,t}$.

4.2 Per-Round Data Design (Original, Synthetic, and Universum) with Worked Example

Our per-round data design follows a separation principle: *evaluate fairness and task loss on original data, while shaping the training distribution using synthetic and Universum points*. Joint (a, y) -stratification with small per-group $y=1$ floors makes each original minibatch statistically representative and stabilizes the EO surrogate, while keeping all EO accounting strictly on originals preserves the mini-to-global guarantees. For the synthetic set, we *fix the label template* (copying (a, y) proportions from the original minibatch, with tiny floors) and *optimize only features* via the bilevel program; this keeps the optimization differentiable and avoids combinatorial label choices. Universum pseudo-positives are then added to counter label imbalance in the training payload, with a capped rule that ties their number to the *synthetic* imbalance but never lets them dominate. Together, these choices yield stable inner gradients, clean fairness evaluation on originals, and privacy preservation—since only synthetic plus Universum features are shared and originals never leave the client.

4.2.1 Minibatch Design on *Original* Data: Notation, Strategy, and Guarantees

Notation. For client k and round t , we use the symbols in Table 2.

Stratified minibatch with floors (original data). Given bounds $B_{\min} \leq |B_{k,t}| \leq B_{\max}$, we construct $B_{k,t}$ as follows:

1. **Choose $|B_{k,t}|$.** Pick $|B_{k,t}|$ large enough to expect p_{target} positives (e.g., $|B_{k,t}| \approx \lceil p_{\text{target}}/\pi_{k,+} \rceil$) while keeping $|B_{k,t}| \in [B_{\min}, B_{\max}]$.
2. **Set quotas.** Start with $q_{k,t}^{(a,y)} \leftarrow \text{round}(\pi_k^{(a,y)} |B_{k,t}|)$. Enforce per-group floors for positives ($y=1$):

$$q_{k,t}^{(a,1)} \leftarrow \max\{q_{k,t}^{(a,1)}, c_{\min,k}\}, \quad a \in \{0, 1\},$$

then adjust the largest remaining cell(s) to preserve $\sum_{a,y} q_{k,t}^{(a,y)} = |B_{k,t}|$.

Symbol	Type/Range	Meaning
$\mathcal{D}_k^o$	set	Client k 's <i>original</i> dataset.
N_k	$\mathbb{N}$	Size of original dataset, $N_k = \mathcal{D}_k^o $.
$a \in \{0, 1\}$	binary	Sensitive attribute.
$y \in \{0, 1\}$	binary	Label (we treat $y=1$ as the minority/positive class).
$N_{k,a,y}^o$	$\mathbb{N}$	# originals in $\mathcal{D}_k^o$ with (a, y) .
$\pi_k^{(a,y)}$	$[0, 1]$	Stratum proportion, $\pi_k^{(a,y)} = N_{k,a,y}^o / N_k$.
$\pi_{k,+}$	$[0, 1]$	Original positive rate, $\pi_{k,+} = (N_{k,0,1}^o + N_{k,1,1}^o) / N_k = \mathbb{P}_k(y=1)$.
$B_{k,t}$	set	<i>Original</i> minibatch; used only for outer loss and EO. Drawn without replacement within the round; reuse across rounds is allowed.
$ B_{k,t} $	$\mathbb{N}$	Size of original minibatch (bounded in practice).
$q_{k,t}^{(a,y)}$	$\mathbb{N}$	Quotas per stratum for $B_{k,t}$; $\sum_{a,y} q_{k,t}^{(a,y)} = B_{k,t} $.
$n_{k,t}^{(y)}$	$\mathbb{N}$	# originals with label y in $B_{k,t}$.
$\Delta_{k,t}$	$\mathbb{N}$	Original-batch imbalance, $\Delta_{k,t} = n_{k,t}^{(0)} - n_{k,t}^{(1)} (\geq 0)$.
$B_{k,t}^{(+)}$	set	Positive slice of $B_{k,t}$; $B_{k,t}^{(+)} = \{(x, a, 1) \in B_{k,t}\}$.
$c_{\min}, c_{\min,k}$	small int	Global (or client-specific) floor on # <i>original</i> positives per sensitive group per minibatch; $c_{\min,k} = \min(c_{\min}, \min_a N_{k,a,1}^o)$.
$\mathcal{D}_{k,t}^s$	set	<i>Labeled synthetic</i> minibatch (labels/attributes fixed; features optimized).
$ \mathcal{D}_{k,t}^s $	$\mathbb{N}$	Size of labeled synthetic minibatch, with $ \mathcal{D}_{k,t}^s \leq B_{k,t} $.
$\tilde{q}_{k,t}^{(a,y)}$	$\mathbb{N}$	Synthetic quotas per stratum; $\sum_{a,y} \tilde{q}_{k,t}^{(a,y)} = \mathcal{D}_{k,t}^s $.
$\Delta_{k,t}^s$	$\mathbb{N}$	Synthetic-batch imbalance, $\Delta_{k,t}^s = \sum_a \tilde{q}_{k,t}^{(a,0)} - \sum_a \tilde{q}_{k,t}^{(a,1)} (\geq 0)$.
$\mathcal{U}_{k,t}^{(Y)}$	set	Universum pseudo-positives ($y=1$), <i>balanced</i> across $a \in \{0, 1\}$.
$ \mathcal{U}_{k,t}^{(Y)} $	$\mathbb{N}$	Universum size this round; see Eq. (7).
$\tilde{S}_{k,t}$	set	Payload to server: $\tilde{S}_{k,t} = \mathcal{D}_{k,t}^s \cup \mathcal{U}_{k,t}^{(Y)}$.

Table 2: Notation for minibatch construction on *original* data and synthetic payloads.

3. **Sample $B_{k,t}$.** Draw $q_{k,t}^{(a,y)}$ points from stratum (a, y) *without replacement* to form $B_{k,t}$. This requires $q_{k,t}^{(a,y)} \leq N_{k,a,y}^o$; we ensure feasibility via $c_{\min,k}$ and the adjustment step. If an exceptionally small stratum would still violate the floor, either (i) temporarily lower that floor in this round and keep sampling without replacement, or (ii) as a last resort, allow *within-batch* replacement for that stratum; unbiasedness claims remain in expectation.

4. **Compute imbalance.** Let $n_{k,t}^{(y)}$ be label counts in $B_{k,t}$ and set $\Delta_{k,t} = n_{k,t}^{(0)} - n_{k,t}^{(1)} \geq 0$.

4.2.2 Design of the *Labeled Synthetic* Minibatch $\mathcal{D}_{k,t}^s$

We fix the *labels* in $\mathcal{D}_{k,t}^s$ by construction and optimize *only* the features via the bilevel program.

1. **Choose $|\mathcal{D}_{k,t}^s|$.** Select a synthetic size with $|\mathcal{D}_{k,t}^s| \leq |B_{k,t}|$.

2. **Set synthetic quotas.** Scale the original quotas to the synthetic size:

$$\tilde{q}_{k,t}^{(a,y)} \leftarrow \text{round}\left(\frac{q_{k,t}^{(a,y)}}{|B_{k,t}|} |\mathcal{D}_{k,t}^s|\right), \quad \sum_{a,y} \tilde{q}_{k,t}^{(a,y)} = |\mathcal{D}_{k,t}^s|.$$

Optionally enforce tiny floors on positives per group (e.g., 2–3) and compensate on the largest $(a, 0)$ cells.

3. **Instantiate label template.** Create $\mathcal{D}_{k,t}^s = \{(x_j^s, a_j^s, y_j^s)\}_{j=1}^{|\mathcal{D}_{k,t}^s|}$ with $\#\{(a_j^s, y_j^s) = (a, y)\} = \tilde{q}_{k,t}^{(a,y)}$. Only $\{x_j^s\}$ are optimized by the outer loop.
4. **Record synthetic imbalance.** Define

$$\Delta_{k,t}^s := \sum_a \tilde{q}_{k,t}^{(a,0)} - \sum_a \tilde{q}_{k,t}^{(a,1)} \quad (\geq 0), \quad \text{so} \quad \Delta_{k,t}^s \leq |\mathcal{D}_{k,t}^s|.$$

4.2.3 Design of the *Universum* Pseudo-Positives $\mathcal{U}_{k,t}^{(Y)}$

Universum points are synthetic $y=1$ examples used to counter label imbalance in the training payload; they do not affect EO evaluation (which is computed on originals). We fix their labels and optimize only their features.

Sensitive balance. Split the sensitive labels in $\mathcal{U}_{k,t}^{(Y)}$ as evenly as possible across $a \in \{0,1\}$ (randomized rounding if $|\mathcal{U}_{k,t}^{(Y)}|$ is odd).

Cardinality rule (driven by synthetic imbalance). We set

$$|\mathcal{U}_{k,t}^{(Y)}| = \min \left\{ \Delta_{k,t}^s, |\mathcal{D}_{k,t}^s|, \Delta_{k,t} \right\}. \quad (7)$$

The first two terms prevent Universum from dominating the synthetic set; the $\Delta_{k,t}$ term ties the payload correction to the imbalance observed in the original minibatch.

Payload. The round- t payload sent to the server is $\tilde{\mathcal{S}}_{k,t} = \mathcal{D}_{k,t}^s \cup \mathcal{U}_{k,t}^{(Y)}$, with labels fixed by the templates above and features optimized by the bilevel. The EO surrogate and outer loss remain computed *only* on the original $B_{k,t}$.

4.2.4 Examples of Minibatch Design (Three Clients)

We give three end-to-end examples (extreme, moderate, and small imbalance). In each case we pick the *original* minibatch size $|B_{k,t}|$, set joint (a, y) quotas with per-group positive floors, compute $\Delta_{k,t}$, build the labeled synthetic set $\mathcal{D}_{k,t}^s$, and determine the Universum count $|\mathcal{U}_{k,t}^{(Y)}|$. We sample *without replacement within each minibatch*; across rounds, reuse is allowed.

Global knobs (used in all scenarios).

- minibatch-size bounds: $B_{\min} = 32$, $B_{\max} = 128$.
- Target positives per original minibatch: $p_{\text{target}} = 8$.
- Floor on positives per sensitive group (original minibatches): $c_{\min} = 5$; feasible per client $c_{\min,k} = \min(c_{\min}, \min_a N_{k,a,1}^o)$.
- Labeled synthetic size (all clients): $|\mathcal{D}_{k,t}^s| = 32$.
- **Synthetic per-group positive floor:** $\tilde{c}_{\min} = 2$. Enforce $\tilde{q}_{k,t}^{(a,1)} \leftarrow \max\{\tilde{q}_{k,t}^{(a,1)}, \tilde{c}_{\min}\}$ and *compensate* by reducing the largest $\tilde{q}_{k,t}^{(a,0)}$ cells so that $\sum_{a,y} \tilde{q}_{k,t}^{(a,y)} = |\mathcal{D}_{k,t}^s|$.

- Universum rule (balanced over a):

$$|\mathcal{U}_{k,t}^{(Y)}| = \min\{\Delta_{k,t}^s, |\mathcal{D}_{k,t}^s|, \Delta_{k,t}\},$$

with a 50/50 split across $a \in \{0, 1\}$ (randomized rounding if odd).

Client 1 (Extreme Imbalance)

Original stratum counts ($N_1=600$):

	$y=0$	$y=1$	Total
$a=0$	300	3	303
$a=1$	270	27	297
Total	570	30	600

Positive rate $\pi_{1,+} = (3 + 27)/600 = 0.05$. Raw size $\hat{B} = \lceil 8/0.05 \rceil = 160$; clamp $\Rightarrow |B_{1,t}| = 128$.

Proportional quotas and floors. Using $\pi^{(a,y)} = N_{1,a,y}^o/600$ and rounding for $|B_{1,t}| = 128$ gives $(64, 1, 58, 6)$. Floor $c_{\min,1} = \min(5, \min\{3, 27\}) = 3$: bump $(a=0, y=1) : 1 \rightarrow 3$ and reduce the largest $y=0$ cell by 2 to keep the sum 128. Final original quotas:

$$(q^{(0,0)}, q^{(0,1)}, q^{(1,0)}, q^{(1,1)}) = (61, 3, 58, 6).$$

Counts: $n_{1,t}^{(1)} = 3 + 6 = 9$, $n_{1,t}^{(0)} = 61 + 58 = 119$, hence $\Delta_{1,t} = 119 - 9 = 110$.

Labeled synthetic quotas ($|\mathcal{D}_{1,t}^s|=32$) with $\tilde{c}_{\min} = 2$. Scale and round:

$$(61, 3, 58, 6) \times \frac{32}{128} \approx (15.25, 0.75, 14.50, 1.50) \Rightarrow (15, 1, 15, 2).$$

Synthetic floor enforcement: $\tilde{c}_{\min} = 2$ requires raising $(a=0, y=1) : 1 \rightarrow 2$. Compensate by reducing the largest $y=0$ cell $(a=0, y=0) : 15 \rightarrow 14$. Final synthetic quotas:

$$(\tilde{q}^{(0,0)}, \tilde{q}^{(0,1)}, \tilde{q}^{(1,0)}, \tilde{q}^{(1,1)}) = (14, 2, 15, 2) \Rightarrow (14, 2, 14, 2) \text{ (adjust largest } y=0 \text{ to keep sum 32)}.$$

Synthetic imbalance:

$$\Delta_{1,t}^s = (14 + 14) - (2 + 2) = 24.$$

Universum (balanced over a).

$$|\mathcal{U}_{1,t}^{(Y)}| = \min\{\Delta_{1,t}^s, |\mathcal{D}_{1,t}^s|, \Delta_{1,t}\} = \min\{24, 32, 110\} = 24 \Rightarrow (12, 12) \text{ across } a.$$

Payload size:

$$|\tilde{S}_{1,t}| = |\mathcal{D}_{1,t}^s| + |\mathcal{U}_{1,t}^{(Y)}| = 32 + 24 = 56.$$

Worst-case enforcement (summary with symbols).

1. **Original floors:** enforce $q_{1,t}^{(a,1)} \leftarrow \max\{q_{1,t}^{(a,1)}, c_{\min,1}\}$, adjust largest $q_{1,t}^{(a,0)}$ to keep $\sum q_{1,t}^{(a,y)} = |B_{1,t}|$.
2. **Synthetic scaling:** $\tilde{q}_{1,t}^{(a,y)} \leftarrow \text{round}(q_{1,t}^{(a,y)} |\mathcal{D}_{1,t}^s| / |B_{1,t}|)$.
3. **Synthetic floors:** enforce $\tilde{q}_{1,t}^{(a,1)} \leftarrow \max\{\tilde{q}_{1,t}^{(a,1)}, \tilde{c}_{\min}\}$ and compensate on the largest $\tilde{q}_{1,t}^{(a,0)}$.
4. **Imbalance:** $\Delta_{1,t}^s = \sum_a \tilde{q}_{1,t}^{(a,0)} - \sum_a \tilde{q}_{1,t}^{(a,1)}$.
5. **Universum:** $|\mathcal{U}_{1,t}^{(Y)}| = \min\{\Delta_{1,t}^s, |\mathcal{D}_{1,t}^s|, \Delta_{1,t}\}$; split 50/50 across a .

Client 2 (Moderate Imbalance)

Original stratum counts ($N_2=900$):

	$y=0$	$y=1$	Total
$a=0$	420	30	450
$a=1$	360	90	450
Total	780	120	900

Positive rate $\pi_{2,+} = 120/900 \approx 0.133$. Raw $\hat{B} = \lceil 8/0.133 \rceil = 60$; choose $|B_{2,t}| = 64$.

Proportional quotas and floors. Proportional counts (30, 2, 26, 6) sum to 64. Floor $c_{\min,2} = \min(5, \min\{30, 90\}) = 5$: bump ($a=0, y=1$) : $2 \rightarrow 5$ and reduce ($a=0, y=0$) : $30 \rightarrow 27$. Final original quotas:

$$(q^{(0,0)}, q^{(0,1)}, q^{(1,0)}, q^{(1,1)}) = (27, 5, 26, 6).$$

Counts: $n_{2,t}^{(1)} = 5 + 6 = 11$, $n_{2,t}^{(0)} = 27 + 26 = 53$, hence $\Delta_{2,t} = 53 - 11 = 42$.

Labeled synthetic quotas ($|\mathcal{D}_{2,t}^s|=32$) with $\tilde{c}_{\min} = 2$. Scale and round:

$$(27, 5, 26, 6) \times \frac{32}{64} = (13.5, 2.5, 13, 3) \Rightarrow (14, 3, 13, 3),$$

sum 33; adjust the largest $y=0$ cell down: (13, 3, 13, 3) (now sum 32). Synthetic floor satisfied (both groups have ≥ 2 positives). Synthetic imbalance:

$$\Delta_{2,t}^s = (13 + 13) - (3 + 3) = 20.$$

Universe (balanced).

$$|\mathcal{U}_{2,t}^{(Y)}| = \min\{20, 32, 42\} = 20 \Rightarrow (10, 10) \text{ across } a.$$

Payload size: $|\tilde{S}_{2,t}| = 32 + 20 = 52$.

Client 3 (Small Imbalance)

Original stratum counts ($N_3=640$; near-balanced):

	$y=0$	$y=1$	Total
$a=0$	160	160	320
$a=1$	160	160	320
Total	320	320	640

Positive rate $\pi_{3,+} = 0.5$. Raw $\hat{B} = \lceil 8/0.5 \rceil = 16$; clamp $\Rightarrow |B_{3,t}| = 32$.

Proportional quotas and floors. For $|B_{3,t}| = 32$, (8, 8, 8, 8) already satisfies $c_{\min,3} = 5$. Thus $\Delta_{3,t} = 16 - 16 = 0$.

Labeled synthetic quotas ($|\mathcal{D}_{3,t}^s|=32$) with $\tilde{c}_{\min} = 2$. Scaling preserves (8, 8, 8, 8); synthetic floor is satisfied and $\Delta_{3,t}^s = 0$.

Universe. $|\mathcal{U}_{3,t}^{(Y)}| = \min\{0, 32, 0\} = 0$; payload size $|\tilde{S}_{3,t}| = 32$.